Degree-Aware Schedules and Consistency Certificates for Subexponential 3-SAT Contraction

Recent-commit review, exact quantitative improvements, and constraints on a change of exponent order

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Abstract

We review the twenty-nine filed refinement packets and five newer incoming packets at a fixed ProveIt commit, together with the user-specified Anthropic formalization. Two quantitative improvements are distinguished. First, the known square-root-degree 3-SAT grouping permits a new strict recursion schedule with eventual time coefficient 13.690 and simultaneous space coefficient 12.798, on the normalized wire scale $s/\sqrt{\lceil \log_2 s \rceil}$. This is a degree-sensitive composition of existing primitives, with the full child-entry cost retained. Its time gain does not require the new consistency inequality.

Second, under the generic root-degree promise $s^{9/10}$, the smallest comparable incoming cached time coefficient is $29.81160894538\ldots$ on the scale $s/\sqrt{\lceil \log_2 s \rceil}$, where s is the normalized Boolean wire count. We improve the certificate for the same finite family of contraction plans by requiring the support chosen for one plan to remain consistent across every pair involving that plan. Conservative pairwise maxima are replaced by exact zero-sum potentials. The resulting time coefficient is $29.74605709379457\ldots$, with simultaneous space coefficient $28.32957818456625\ldots$, certified by integer arithmetic and an independently reconstructed checker. The complete recursion charges every child-table entry and preserves the inherited degree and precision contracts. These are improvements of exponential upper-bound coefficients. They do not establish a smaller logarithmic order for general 3-SAT, and the algorithms use subexponential space.

We also prove a full-rank obstruction to a universal linear summary of variable unions, and sharpen the counting of degree arrays in an already known grouping reduction. These results identify specific limits of two routes to a larger improvement without asserting a lower bound on 3-SAT algorithms. The archive contains complete LaTeX sources, exact certificate data, independent checks, and a reproducible finite contraction prototype.

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1 Outcome and the comparison actually being made

There are two results. The one relevant to the known arbitrary-density 3-SAT regrouping uses its stronger root-degree promise $\Delta \le s^{51/100}$. A new strict degree-aware schedule gives the eventual simultaneous bounds

$$T \le (s+\mathcal{L})^{O(1)} 2^{13.690 s/\sqrt{\lceil \log_2 s \rceil}}, \qquad S \le (s+\mathcal{L})^{O(1)} 2^{12.798 s/\sqrt{\lceil \log_2 s \rceil}}.$$

Section 5 proves this composition and distinguishes its time improvement from the new support-consistency argument. The latter also improves the wider generic interface: under the same root-degree promise $\Delta \le \lceil s^{9/10} \rceil$ as the strongest incoming cached packet, it gives

$$\begin{aligned} T(s,\mathcal{L}) &\le (s+\mathcal{L})^{O(1)} 2^{(29.74605709379457\ldots+o(1))s/\sqrt{\lceil \log_2 s \rceil}}, \\ S(s,\mathcal{L}) &\le (s+\mathcal{L})^{O(1)} 2^{(28.32957818456625\ldots+o(1))s/\sqrt{\lceil \log_2 s \rceil}}. \end{aligned} \tag{1}$$

Here $\mathcal{L}$ is the description length, and the two bounds are a simultaneously valid pair. The theorem is relative to explicit imported deformation, recovery, packing, and matrix multiplication results. The new finite inequality and its exact certificate are proved and checked here.

The request for a breakthrough in the unrestricted logarithmic order is *not resolved*. In particular, this work does not replace $n \log\log n/\sqrt{\log n}$ by $n/\sqrt{\log n}$ or by $n \log\log n/\log n$. The improvement is nevertheless asymptotic at the level of the displayed quantitative upper bounds within the specified degree and resource classes. The generic coefficient alone would not improve the strongest contractor usable after the existing 3-SAT regrouping; the lower-degree specialization is essential. Ratios of these upper bounds are not lower bounds on the running times of implementations.

1.1 A frozen review, including the latest arrivals

The source snapshot is ProveIt commit `8e76d515063c4b9fbd8743a846b457e3a01bc039`. It includes the five archives added by `26bd83d4b37d` at 8:51 PM Pacific on October 26, 2026. The preceding placement `a7ae5ab55f96` moved nine fourth-round packets into the research directory, and `7de9d09fecd1` linked the public Anthropic development. The repository-wide head also contains unrelated work; our mathematical review concerns the two requested directories and the referenced source manuscripts only [1, 2, 3].

We obtained the report sources and headline/status files for all twenty-nine filed refinement packets, then extracted and inspected all five new archives. The downloaded source inventory contains 148 objects, each verified against its advertised Git blob identifier. Detailed proof review concentrates on the strongest compatible bounds and the mathematical interfaces used below. This is not a claim that every proof in thirty-four packets has been independently audited. Appendix A records the scope.

The five arrivals all retain the same general SAT exponent order. Their generic degree-$s^{9/10}$ cached resource bounds are:

Incoming packet	Time coefficient	Space coefficient
`sat_affine_cauchy`	30.413349	29.466030
`sat_affine_contraction`	30.257036	25.989068
`sat_affine_geometry`	30.269873	29.487537
`sat_affine_plane`	30.691338	29.466030
`sat_pairwise_certificates`	29.811609	29.811609
This report	29.746057	28.329578

All coefficients multiply $s/\sqrt{\log_2 s}$, equivalently $s/\sqrt{\lceil \log_2 s \rceil}$ to leading order. All rows use subexponential space. The smaller-memory affine construction remains a separate option. The best reviewed generic polynomial-space time coefficient is 46.18484558673…; the separate degree-$s^{1/12}$ polynomial-space coefficient is 16.81223380188… [10, 13, 14]. These stronger degree assumptions must not be silently substituted into the generic comparison.

Conversely, a generic bound must not be advertised as best for 3-SAT when its reduction already produces lower degrees. The earlier exact-kernels packet states a 25.16377… polynomial-space coefficient at degree $s^{0.51}$. The earlier eight-group cached construction has a 19.08387… branch whose proof in fact extends to this same degree class, even though its stated corollary assumes $s^{1/12}$. The new 13.690 coefficient is compared with these in Section 5; it uses subexponential space. The already available lower-degree reductions are credited in Section 7.

1.2 The two source manuscripts and their formal scope

For 3-SAT, the OpenAI October 25 manuscript gives the general order

$$(n+m)^{O(1)}2^{O(n\log\log n/\sqrt{\log n})}$$

in polynomial space, and the fixed-density order $2^{O_D(n/\sqrt{\log n})}$ when $m \le Dn$, with polynomial input factors understood. Its pinned Lean scope page explicitly describes the weaker formalized exponent $n(\log\log n)^2/\sqrt{\log n}$, still with polynomial space. The manuscript's sharper appendix and the formalized scope are different guarantees [4, 5].

At the exact Anthropic commit supplied by the user, `1a6eee293b9e`, `MainResults.MainTheorem` states the existence of one deterministic finite-stack Mathlib machine and a constant C that count all satisfying assignments within

$$2^{C\min\{3m/\sqrt{\log_2 m},\, 9n\log_2\log_2 n/\sqrt{\log_2 n}\}}$$

on the stated 3-CNF inputs. Those inputs have $n \ge 4$, $m \ge 2$, no duplicate clause lists, and every declared variable occurring. Every stack alphabet is explicitly required to be finite. The proof entry point supplies a term for this proposition. The headline proposition contains no polynomial-space conclusion [6, 7].

We inspected these statements and proof entry points; we did not rebuild or independently replay either large Lean development. This report neither denies the formal statements nor claims a fresh kernel verification. Unknown constants in the separate Anthropic construction do not permit a numerical speed comparison with the normalized-wire coefficient in (1). Our claims concern explicit normalized contraction bounds in their stated degree classes, not a proof of being faster than every possible implementation of both manuscripts.

1.3 What the new idea changes

The strongest incoming certificate bounds the norm of an average of nine support vectors by summing thirty-six separately maximized pair terms. The support chosen for one plan can consequently be different in each pairwise maximum. A real simultaneous selection has only one support for that plan. We exploit this consistency by redistributing zero-sum functions of each selected support among its incident pairs.

This preserves the contraction plans, arithmetic backend, and recursion. The simultaneous storage bound adds a boundary-size objective to the exact plan selector. The primary improvement is in the

proof of the worst-case running time. Pairwise consistency and such potential reparametrizations are classical finite optimization tools [21]; the contribution here is their exact certificate for this specific contraction family and the fully charged consequence.

2 Exact contraction interface and local cost supports

2.1 Input promises and imported dependencies

The normalized input is a bipartite Boolean tensor network with at most s wires, polynomially many vertices, and maximum degree at most $\lceil s^{9/10} \rceil$. Its value is

$$Z = \sum_{x \in \{0,1\}^E} \prod_v T_v(x|_{\delta(v)}).$$

The entries belong to $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$ with $1 \le P_0, \ell \le s$. A uniform entry procedure uses polynomial space and $(s + \mathcal{L})^{O(1)} 2^{O(s^{9/10})}$ bit operations, with fixed constants. All subsequent estimates use these same promises.

The inherited computation extends scalars to

$$R_p = (\mathbb{Z}/2^p\mathbb{Z})[U, V]/(f(U), V^\ell),$$

where f is monic of degree $O(\log s)$ and irreducible modulo two. Ring rank and coordinate precision remain polynomial in s. A tensor description retains its creation precision and is evaluated using the same canonical coordinate representatives at later precisions. Canonical lifting is not used as a homomorphism.

We import the exact Cauchy deformation, local integrality, Hermite recovery, and original SAT reductions from the source, and the compatible refinements of packing, full-orientation modes, and sharp local precision from the reviewed packets [4, 8, 9, 10, 13]. The proof below starts after those constructions have produced finite mode lists. It only changes how the ensuing finite sum is evaluated and bounded.

2.2 The mode sum and its entropy inequality

At one active call, let M be the wire count, r a fixed group count, u the retained valuation range, a the packet exponent, and $D = 2^{a-i}$ for $0 \le i < u$. The imported identities give

$$p_t \le (6 + o(1)) \frac{MD}{u}, \qquad M_{\text{child}} \le \left(\frac{2}{r} + o(1) \right) M.$$

For each directed group pair (g, h), $g \ne h$, enumerate its exact valuation budget and let $\mathcal{I}_{gh}$ be the finite list of mode exponent tuples realizing that budget. Empty lists contribute zero and are skipped. With r fixed, the budget arrays are polynomially numerous in s. Their exact list cardinalities, ranking, and unranking are supplied by the ordinary finite knapsack recurrence for the known integral mode weights.

For one budget array the contribution is precisely

$$\sum_{\xi_{gh} \in \mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}). \tag{2}$$

An entry of Z_g is an actual recursive contraction, with its cost charged later. Write

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \quad g_{\{g,h\}} = h_{gh} + h_{hg}, \quad K(M,u) = \kappa \frac{M}{\sqrt{u}}, \qquad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

The inherited common-tilt estimate states that for each fixed nonnegative unordered support v,

$$v \cdot g \le \left(\frac{\sqrt{2}\|v\|_2}{r} + o(1) \right) K(M,u). \tag{3}$$

The error is uniform on a fixed finite support family. The factor $\sqrt{2}$ converts an unordered edge into its two directed copies and must be retained.

For orientation, the mode weights have the form $a + 1 - i + 4j - 2\,\mathrm{popcount}(j)$, and at most $(2/r^2 + o(1))M/D$ modes of each index occur on a directed pair. The generating product has logarithm $\pi^2/(24\theta) + O(\log^2(2/\theta))$ as $\theta \downarrow 0$. Combining this with the common precision budget and minimizing in θ gives (3). The buffers in Section 4 make its error uniform on all active calls. We use this already proved interface, rather than asserting that a finite checker proves its analytic asymptotics.

2.3 Contraction plans and the arithmetic backend

A plan is an ordered partition of $[r]$ into three nonempty blocks. Each block is built recursively by splitting its ordered list at its midpoint. Merging disjoint blocks A, B with outside block O is a matrix product whose three index classes correspond to AB, AO, BO. After the three main blocks have been built, the remaining triangle is contracted. Every step follows from finite distributivity over R_p.

The retained rectangular backend uses fixed field bounds $\omega < 19/8$ and dual exponent $\alpha > 3/10$. These weaker rational allowances follow from the published bounds in [20]. For sorted logarithmic dimensions $x \ge y \ge z \ge 0$, set $\varepsilon = 10^{-5}$. The reviewed implementation gives the exponent

$$F(x,y,z) = \max\left\{(1+\varepsilon/2)(x+y), \quad x + (47/56+\varepsilon)y + (15/28)z\right\}. \tag{4}$$

The input/output term is included. As a function of the unsorted triple, this is the maximum of the distinct permutations of $(1+\varepsilon/2, 1+\varepsilon/2, 0)$ and $(1, 47/56+\varepsilon, 15/28)$, by the rearrangement inequality.

The exact ring bridge lifts coordinates to bounded integers, uses finitely many nonexceptional odd primes for the fixed rational bilinear algorithms, reconstructs integer products by CRT, and reduces modulo 2^p after combining the ring structure constants. The number of coordinate products, primes, and bits per coordinate is polynomial in s. Thus the field algorithms yield the same displayed exponential coefficient over R_p without inverting a nonunit. A fixed bilinear recursion processes one recursive product at a time; its working storage is a polynomial factor times the largest relevant input/output table [12].

Put $L_* = 1{,}400{,}000$. The integral triples are

$$(1400007, 1400007, 0), \qquad (1400000, 1175014, 750000). \tag{5}$$

An operation produces nine vectors by assigning these permuted triples to its three crossing edge classes and zero to edges internal to a class. There are $r-2$ operations per plan. At $r = 10$ this

gives 72 supports in dimension $\binom{10}{2} = 45$. Denote their multiset by $\mathcal{V}_j$. The time score for plan j is

$$F_j(g) = \frac{1}{L_*} \max_{v \in \mathcal{V}_j} v \cdot g.$$

Repeated supports are harmless; their deterministic order is fixed by postorder traversal and lexicographic order of the nine triples.

For each nonempty stored block A, let b_A be the indicator of its edge boundary. There are 17 distinct such vectors per ten-group plan, including all singleton tables. Define the storage score

$$G_j(g) = \max_A b_A \cdot g. \tag{6}$$

Every input and output of an internal matrix product is one of these boundary tables. The final three matrices are also included. Constantly many simultaneous tables change their size only by a constant factor.

2.4 Exact plan selection

The final algorithm chooses a plan minimizing

$$J_j(g) = \max\{F_j(g), (21/20)G_j(g)\}. \tag{7}$$

This one choice will achieve both the time and space bounds. It is not legitimate to choose one plan for time and another plan for space and combine their guarantees.

There is an exact integer implementation. Set $d_{gh} = |\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|$ for unordered pairs. Augment each time support by the integer vectors $1{,}470{,}000\, b_A$. For every augmented support v, form $\prod_e d_e^{v_e}$, take the maximum within a plan, and choose the plan of smallest maximum, breaking ties by index. Taking a base-two logarithm shows equivalence to (7). The exponents, group count, and menu size are fixed constants; the integers have polynomial bit length in s. No floating-point comparison or optimization oracle is needed at runtime.

3 A certificate that preserves support consistency

3.1 A general exact inequality

Let $\mathcal{W}_1, \dots, \mathcal{W}_q$ be nonempty finite indexed families of vectors in $\mathbb{R}^d$, and let $p_i \geq 0$, $\sum_i p_i = 1$, $q \geq 2$. Identical vectors may have distinct indices and distinct potentials. For $i < j$ define

$$c_{ij}(v, z) = \frac{p_i^2 \|v\|_2^2 + p_j^2 \|z\|_2^2}{q - 1} + 2 p_i p_j \langle v, z \rangle.$$

Choose a function $\phi_{ij,i}$ on $\mathcal{W}_i$ and $\phi_{ij,j}$ on $\mathcal{W}_j$ for each unordered pair. Require

$$\sum_{j \neq i} \phi_{\{i,j\},i}(v) = 0 \qquad (i \in [q],\ v \in \mathcal{W}_i). \tag{8}$$

Set

$$A_{ij} = \max_{v \in \mathcal{W}_i, z \in \mathcal{W}_j} \{c_{ij}(v, z) + \phi_{ij,i}(v) + \phi_{ij,j}(z)\}, \qquad U = \sum_{i<j} A_{ij}. \tag{9}$$

Theorem 3.1 (Consistency certificate)**.** *For every simultaneous choice $v_i \in \mathcal{W}_i$,*

$$\|\sum_i p_i v_i\|_2^2 \le U.$$

If every vector is nonnegative and the entropy estimate (3) applies, the plan minimizing $\max_{v\in\mathcal{W}_i} v\cdot g$ has local coefficient at most $\beta = \sqrt{2U}/r$.

Proof. Each squared term $p_i^2\|v_i\|_2^2$ occurs in $q-1$ pair costs, with coefficient $1/(q-1)$. Thus

$$\|\sum_i p_i v_i\|_2^2 = \sum_{i<j} c_{ij}(v_i, v_j) = \sum_{i<j}\{c_{ij}(v_i, v_j) + \phi_{ij,i}(v_i) + \phi_{ij,j}(v_j)\}.$$

The second equality is exactly (8); it uses the same v_i at every pair containing i. Bound each summand by its maximum to obtain the first conclusion.

For a given g, choose a support v_i maximizing each plan's score. The minimum of those scores is no larger than their convex average, which is $(\sum_i p_i v_i)\cdot g$. The averaged support is nonnegative and belongs to a fixed finite family. Apply (3) and the first conclusion. No interchange of a minimum and maximum, or simultaneous realizability of arbitrary pair maximizers, is asserted. □

Zero potentials recover the incoming pair certificate. Nonzero potentials can decrease the sum of maxima while leaving every true simultaneous quadratic expression unchanged. Negative potential values are permitted; only the support vectors used in the entropy estimate need to be nonnegative.

3.2 The fixed ten-group family and exact data

We retain the incoming nine ordered plans. Their new proof weights have denominator $W = 100000$:

j	w_j	First block	Second block	Third block
0	12527	$(7,4,3,9)$	$(1,6,0)$	$(8,2,5)$
1	11761	$(5,4,6,7)$	$(9,1,8)$	$(2,3,0)$
2	13353	$(0,5,9,7)$	$(2,1,4)$	$(8,3,6)$
3	11841	$(0,4,7,8)$	$(2,9,6)$	$(1,5,3)$
4	10132	$(3,1,7,2)$	$(9,6,8)$	$(5,4,0)$
5	12324	$(7,1,5,8)$	$(2,9,4)$	$(0,6,3)$
6	11088	$(7,2,8,6)$	$(5,9,3)$	$(0,1,4)$
7	10184	$(1,9,6,7)$	$(2,5,0)$	$(4,3,8)$
8	6790	$(4,6,1,5)$	$(8,9,2)$	$(0,7,3)$

Use normalized supports v/L_* in the theorem and weights $p_j = w_j/W$. The potentials are integers divided by $P = 10^{12}$. The complete arrays are in `data/support_consistency_certificate.json`.

To state the finite certificate without dependence on an optimizer, put

$$D_* = (q-1)W^2L_*^2 = 156800000000000000000000.$$

For raw integral supports v, z and potential numerators $a_{ij,i}(v)$, the exact pair numerator is

$$N_{ij}(v,z) = w_i^2\|v\|_2^2 + w_j^2\|z\|_2^2 + 16w_iw_j\langle v, z\rangle + \frac{D_*}{P}(a_{ij,i}(v) + a_{ij,j}(z)). \tag{10}$$

D_*/P is an integer. Exhaustively maximizing these integers for each pair and dividing their sum by D_* gives

$$U = \frac{283611964650596186827}{17500000000000000000}, \qquad \boxed{\beta_{10}^2 = \frac{2U}{10^2} = \frac{283611964650596186827}{875000000000000000000}}. \tag{11}$$

There are $9 \cdot 72 = 648$ conservation equations and $\binom{9}{2}72^2 = 186624$ time-support pairs. Their exact maxima and witnesses are recorded with the certificate. The supplied checkers reconstruct the supports from the ordered plans and (5); they do not trust an externally supplied support matrix.

3.3 Joint storage control with the same certificate

For every stored-block boundary b_A, the original time family contains an input/output support $1400007\,b_A$. Append the support $1470000\,b_A$, and give it the same potentials as that input/output support. Conservation remains exact. The new normalized support is $(21/20)b_A$, so the maximum over this augmented family is precisely the score in (7).

Lemma 3.2 (Augmented certificate)**.** *For the fixed data above, adding the* 17 *boundary supports per plan does not increase any exact pair maximum. Therefore the selector* (7) *simultaneously guarantees*

$$F_j(g) \le (\beta_{10} + o(1))K(M,u), \qquad G_j(g) \le \left(\frac{20}{21}\beta_{10} + o(1)\right) K(M,u).$$

Proof. There are 89 augmented supports per plan. The time–time choices have already been checked. The remaining number of choices is

$$\binom{9}{2}(89^2 - 72^2) = 98532.$$

For each such choice the integer expression (10) is at most the corresponding old maximum. The strict minimum margin is positive and is reproduced by both exact checkers; no numerical comparison is used. This finite inequality is the certificate premise of Theorem 3.1. There are 801 conservation equations in the enlarged family. Apply that theorem to the augmented score, then use the two terms in its defining maximum. □

3.4 Discovery and its separation from verification

The discovery method is a finite potential redistribution. With current corrected pair matrix $M_{ij}(v,z)$, fix plan i and define

$$a_{ij}(v) = \max_z M_{ij}(v,z), \qquad \overline{a}_i(v) = \frac{1}{q-1}\sum_{j\neq i} a_{ij}(v).$$

Update $\phi_{ij,i}(v)$ by adding $\overline{a}_i(v) - a_{ij}(v)$. The increments sum to zero for every v. Each incident pair's conditional maximum then becomes $\overline{a}_i(v)$, and the sum of its unconditional maxima cannot increase, because

$$(q-1)\max_v \overline{a}_i(v) \le \sum_{j\neq i} \max_v a_{ij}(v).$$

This supplies a simple descent procedure. Numerical sweeps and a short search over proof weights suggested the displayed data.

For each plan, the potentials on all but its last incident pair were rounded to denominator 10^{12}. The final potential was set to the negative exact sum of the preceding ones. All pair maxima were then recomputed as integers. The accepted certificate needs no floating-point tolerance, optimization software, convergence assertion, or optimality claim about the discovery procedure.

3.5 A nearly sharp limit for this particular proof method

Let t_i be the raw support with index $67, 67, 69, 69, 67, 69, 67, 69, 69$, respectively, in the nine plans. Put $G_{ij} = \langle t_i, t_j \rangle$. For any mixture p with $\sum_i p_i = 1$, this one allowed simultaneous tuple already has squared normalized norm $p^T G p / L_*^2$. The exact checker verifies that G is positive definite and $G^{-1}\mathbf{1}$ has positive coordinates. Completing the square gives

$$\frac{p^T G p}{L_*^2} \geq U_* := \frac{1}{L_*^2 \mathbf{1}^T G^{-1} \mathbf{1}} = 16.206397979569626\ldots. \tag{12}$$

The corresponding coefficient lower bound at the fixed root taper lies within $4.3 \cdot 10^{-10}$ of the coefficient proved here. The comparison is certified with rational bounds, including the exact Gram matrix and its positive-definiteness certificate.

Thus another search over the mixture weights of these same nine plans cannot yield a material improvement *through this worst-case average-norm argument.* This does not establish optimality of the actual adaptive plan selector. Different menus, a proof of the minimum of plan costs that avoids a fixed average, stronger information on generated mode lists, or a different contraction representation remain possible.

4 The complete paid recursion

4.1 Every child-table entry has a cost

For a group count r, a singleton factor is incident with $2(r-1)$ directed pair lists. Applying the common entropy inequality to their indicator gives the entry-count coefficient

$$\alpha_r = \frac{\sqrt{2(r-1)}}{r}.$$

Filling a singleton table invokes one recursive child contraction for every entry. A locally certified aggregation coefficient β_r therefore gives the paid recurrence

$$T_d \leq \mathrm{poly}(s) \left[2^{(\alpha_{r_d} + o(1))K_d} T_{d+1} + 2^{(\beta_{r_d} + o(1))K_d} \right], \qquad K_d = \kappa \frac{M_d}{\sqrt{u_d}}. \tag{13}$$

Using only the smaller local aggregation term would omit the main recursive work. We retain the complete recurrence.

The root uses the joint certificate of Lemma 3.2. At depth one we retain the incoming thirteen-group three-plan certificate $\beta_{13}^2 = 1572/4225$. In the tail we use the seven-group plan $(0,1,2) \mid (3,4) \mid (5,6)$ with $\beta_7^2 = 632/1225$. Those two local estimates use the previously reviewed $12/5$ matrix allowance and have the same entry and storage conventions [11, 13]. Their support checks are included in our replay.

4.2 An explicit integer schedule

Write $\lambda = \lceil \log_2 s \rceil$, $B = \lceil 6 \log_2 \lambda \rceil$. Keep the preceding fastest packet's schedule:

$$\begin{aligned} r_0 &= 10, & \eta_0 &= \frac{301}{20000}, \\ r_1 &= 13, & \eta_1 &= \frac{119}{10000}, \\ r_d &= 7 \quad (d \geq 2), & \eta_d &= \frac{1549}{324000}\left(\frac{4}{9}\right)^{d-2}. \end{aligned} \tag{14}$$

Set

$$u_d = \lfloor \eta_d \lambda \rfloor, \quad a_d = \sum_{j=0}^{d} (u_j + B), \quad L_d = 2^{a_d}, \quad Q_d = r_d^{a_d}.$$

The cumulative retained fractions are

$$\eta_0 + \eta_1 = \frac{539}{20000}, \qquad \sum_{d \geq 0} \eta_d = \frac{8}{225}.$$

Use padding with relative slack λ^{-3} and stop at $M \leq H = \lceil s/\lambda^{3/4} \rceil$. For a finite exceptional range of s, use direct evaluation. This is an eventual asymptotic certificate; we do not assert that its small positive gate margins yield a useful finite-size crossover.

4.3 Degree, packing, and depth gates

With side degrees bounded by R, C, the sufficient inherited packing gates are

$$QR/M = o(1), \qquad Q^2 C (\log s)^2 / M = o(1), \qquad LQ^2/M = o(1). \tag{15}$$

Initially $R, C \leq s^{9/10+o(1)}$. Subsequently the retained side stays under that bound and the fresh side has degree at most $2^{a_{d-1}+1}$.

The exact integer inequalities

$$10^{1000} < 2^{3322}, \qquad 13^{1000} < 2^{3701}, \qquad 7^{100} < 2^{281}$$

give strict rational upper bounds on the relevant logarithms. The largest old-degree exponents are consequently separated from one by

Stage	Exponent upper bound	Positive gap
Root	$9/10 + 2\eta_0(3322/1000)$	$39/5000000$
Depth one	$9/10 + (539/20000)(3701/1000)$	$5161/20000000$
Tail	$9/10 + (8/225)(281/100)$	$1/11250$

The fresh-side and packet-capacity exponents are at most

$$\frac{8}{225}\left(1 + 2 \cdot \frac{3701}{1000}\right) < 1.$$

All gaps are fixed positive constants. They absorb the additive $O((\log \lambda)^2)$ guards, all fixed polylogarithmic factors, and $s/M \leq \lambda^{3/4}$ on active calls. Fresh degrees remain below $s^{8/225+o(1)} < s^{9/10}$.

For a noncircular depth argument, provisionally consider depths through $\lceil (3/4)\log_3 \lambda \rceil + O(1)$. Since $4/9 > 1/3$, the smallest tail taper in that range is at least $\lambda^{1/4+o(1)}$. The gates just proved therefore hold uniformly. Nearly lossless padding then makes each tail child smaller than $M/3$ for all sufficiently large s. An active path cannot exceed the provisional depth. This closes the induction, proves all active tapers diverge, and makes their floors negligible. The padding errors over $O(\log \lambda)$ levels multiply to $1 + o(1)$, yielding

$$M_0 \le s, \quad M_1 \le (1+o(1))s/5, \quad M_d \le (1+o(1))\frac{2s}{65}(2/7)^{d-2} \quad (d \ge 2). \tag{16}$$

4.4 Precision, entry access, and leaves

For a retained valuation $i < u_d$, $D = 2^{a_d - i} \ge 2^{a_{d-1}+B+1}$. The inherited creation-time precision bound is $p = O(s\lambda^2 2^{a_{d-1}})$. Hence

$$\frac{p}{MD/u_d} = O(\lambda^{15/4} 2^{-B}) = o(1).$$

The additional Hermite terms are absorbed by the same buffer. The local precision bound and entropy estimate thus apply uniformly, while $a_d \le (8/225)\lambda + O((\log \lambda)^2)$ keeps every ring rank and precision polynomial in s.

Original entry routines have cost $2^{O(s^{9/10})}$ times a polynomial; fresh packet entry routines have cost $\exp(s^{8/225+o(1)})$. Direct leaves cost $2^{O(s/\lambda^{3/4})}$ times these entry costs. Each is lower order on the scale $s/\sqrt{\lambda}$. The immutable entry descriptions prevent expansion of an entire ancestor contraction inside an entry call. We first count arithmetic and entry requests, then charge their uniform description-access cost; the input-length polynomial is paid once. Polynomial overhead in s over $O(\log \lambda)$ levels adds $O(\lambda \log \lambda) = o(s/\sqrt{\lambda})$ to the exponent.

4.5 Unrolling every time branch and the simultaneous space bound

On the normalized scale define

$$K_0^* = \frac{\kappa}{\sqrt{\eta_0}}, \qquad K_1^* = \frac{\kappa}{5\sqrt{\eta_1}}, \qquad K_2^* = \frac{2\kappa}{65\sqrt{\eta_2}}.$$

The tail ratio is $\gamma = (2/7)/\sqrt{4/9} = 3/7$. Its entry sum has coefficient $\alpha_7/(1-\gamma) = \sqrt{3}/2 > \beta_7$. A tail aggregation at offset t is bounded by a convex combination of the initial tail aggregation and the infinite entry sum. Therefore the only candidates needed to bound every branch of (13) are

$$C_0 = \beta_{10} K_0^* = \frac{\pi}{\ln 2}\sqrt{\frac{2\beta_{10}^2}{301/20000}} = 29.74605709379457\ldots, \tag{17}$$

$$C_1 = \alpha_{10} K_0^* + \beta_{13} K_1^* = \frac{\pi}{\ln 2}\left(\sqrt{\frac{7200}{301}} + \sqrt{\frac{50304}{20111}}\right) = 29.3352039993294\ldots, \tag{18}$$

$$\begin{aligned} C_\infty &= \alpha_{10} K_0^* + \alpha_{13} K_1^* + \frac{\alpha_7 K_2^*}{1-\gamma} \\ &= \frac{\pi}{\ln 2}\left(\sqrt{\frac{7200}{301}} + \sqrt{\frac{19200}{20111}} + \sqrt{\frac{77760}{261781}}\right) = 29.065748524641\ldots. \end{aligned} \tag{19}$$

Rational interval arithmetic verifies $C_0 > C_1, C_\infty$ and $C_0 < 29.747$. At the root the joint selector bounds local storage by $(20/21)C_0$. At subsequent depths the local storage coefficients are at most

7.16817894875 and 2.04877062910, followed by a decreasing geometric tail. Caches on one active recursion path coexist, so their *sizes add*: their leading exponent is the maximum, not the sum.

Theorem 4.1 (Improved normalized contraction bound)**.** *Under the input promises and imported exact interfaces of Section 2, the described deterministic contraction procedure has simultaneous bounds*

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_T+o(1))s/\sqrt{\lceil \log_2 s\rceil}},$$
$$S(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)} 2^{(C_S+o(1))s/\sqrt{\lceil \log_2 s\rceil}},$$

where

$$\boxed{C_T = 29.74605709379457\ldots, \qquad C_S = \frac{20}{21}C_T = 28.32957818456625\ldots.}$$

The safe eventual coefficients 29.747 *and* 28.330 *are valid.*

Proof. The exact plan changes only the association and order of the finite sums and products in (2). The imported identity and recovery therefore give the original contraction. The gate and depth argument justifies every call. The finite certificates and the paid recurrence give the three time candidates, whose maximum is C_0. The simultaneous storage argument gives C_S. All remaining costs and uniform errors have been charged above. Strict gaps to the displayed safe coefficients absorb the lower-order terms for sufficiently large s. □

5 A degree-aware specialization for the 3-SAT reduction

The generic degree allowance $s^{9/10}$ is deliberately broad. A reduction with a smaller maximum degree should use a different retained-range schedule. Here we prove a separate bound under maximum initial degree $\lceil s^{51/100}\rceil$. Section 6 records the wire and degree premises needed to apply such a contraction theorem to 3-SAT. The new schedule is useful in its own right; it is not obtained by simply rescaling the generic displayed coefficient.

5.1 A strict rational schedule

Retain the same group sequence $10, 13, 7, 7, \ldots$ and local coefficients as in Section 4. Put

$$\eta_0 = \frac{59}{800}, \qquad \eta_1 = \frac{181}{4000}, \qquad \eta_d = \frac{89}{5000}\left(\frac{4}{9}\right)^{d-2} \quad (d \ge 2).$$

For $\lambda = \lceil \log_2 s\rceil$, use

$$u_d = \lfloor \eta_d \lambda \rfloor, \qquad B = \lceil 6\log_2 \lambda\rceil, \qquad a_d = \sum_{j\le d}(u_j + B),$$

relative padding slack λ^{-3}, and direct evaluation below $H = \lceil s/\lambda^{3/4}\rceil$. The two relevant cumulative fractions are

$$A_1 = \eta_0 + \eta_1 = \frac{119}{1000}, \qquad A_\infty = \eta_0 + \eta_1 + \frac{9}{5}\eta_2 = \frac{472}{3125}.$$

The fixed logarithm bounds

$$\log_2 10 < \frac{1661}{500}, \qquad \log_2 13 < \frac{3701}{1000}, \qquad \log_2 7 < \frac{281}{100}$$

are certified by integer powering in the supplied checker. The following positive gaps are exact rational values. The degree column subtracts $51/100+2\eta_0\log_2 10$ at the root and $51/100+A_d\log_2 r_d$ afterwards from one. The capacity column subtracts $(1+2\log_2 r_d)A_d$ from one.

Stage	Degree gap	Capacity gap
Root, $r=10$	1/200000	87251/200000
Depth one, $r=13$	49581/1000000	81/500000
Tail, $r=7$	20493/312500	9/78125

These are bounds using the displayed rational logarithm upper bounds. The fresh-side gate has exponent $2A_d\log_2 r_d+A_{d-1}\le(1+2\log_2 r_d)A_d$, so the capacity column also pays that gate. Moreover $A_\infty<51/100$, ensuring that all newly created packet degrees remain within the retained degree budget. This proof uses stage-specific capacity bounds; replacing every tail logarithm by $\log_2 13$ would unnecessarily discard this schedule.

For completeness, the depth and precision arguments still close uniformly. Provisionally consider at most $\frac{3}{4}\log_3\lambda+O(1)$ active tail levels. The smallest retained range in that window is a positive power of λ, since $(9/4)^3<3^4$. All fixed exponent gaps in the table absorb the $O((\log\lambda)^2)$ cumulative precision buffers and $s/M\le\lambda^{3/4}$. Nearly lossless packing then gives actual tail child ratio $2/7+o(1)<1/3$, proving the provisional depth bound. The product of all relative padding losses is $1+o(1)$, and floors are uniformly negligible. The retained deformation scales satisfy $D\ge 2^{a_{d-1}+B+1}$. Consequently the imported creation precision $O(s\lambda^2 2^{a_{d-1}})$ divided by MD/u_d is $O(\lambda^{15/4}2^{-B})=o(1)$. This justifies the same sharp precision and entropy bounds used in the generic theorem.

All numerical representations have polynomial bit length, because $a_d\le(472/3125)\lambda+O((\log\lambda)^2)$. The unchanged entry-access costs and the direct leaves are lower order on $s/\sqrt{\lambda}$: the generic permitted original-entry cost $2^{O(s^{9/10})}$ already suffices, while new packet descriptions cost $\exp(s^{472/3125+o(1)})$ and leaves cost $2^{O(s/\lambda^{3/4})}$. No stronger entry-access assumption is silently introduced by the smaller degree premise.

5.2 Charge the recurrence before comparing coefficients

Let β^2 be the new exact ten-group coefficient from Section 3. The retained thirteen- and seven-group coefficients remain $\beta_{13}^2=1572/4225$ and $\beta_7^2=632/1225$, with $\alpha_r=\sqrt{2(r-1)}/r$. The geometric tail ratio is again 3/7. Unrolling the paid recurrence gives

$$C_0=\frac{\pi}{\ln 2}\sqrt{\frac{1600\beta^2}{59}}=13.43744330254181\ldots,$$
$$C_1=\frac{\pi}{\ln 2}\left(\sqrt{\frac{288}{59}}+\sqrt{\frac{100608}{152945}}\right)=13.68968074357885\ldots,$$
$$C_\infty=\frac{\pi}{\ln 2}\left(\sqrt{\frac{288}{59}}+\sqrt{\frac{7680}{30589}}+\sqrt{\frac{1200}{15041}}\right)=13.56492576337809\ldots.$$

Every tail local branch lies below the terminal branch because $\alpha_7^2/(1-3/7)^2=3/4>\beta_7^2$. Exact rational enclosures prove $C_1>\max(C_0,C_\infty)$.

The joint root selector gives local root space coefficient $\frac{20}{21}C_0=12.79756505003982\ldots$. The depth-one and initial tail coefficients are respectively $3.67597904980247\ldots$ and $1.06178489349021\ldots$; later

tail coefficients decrease geometrically. Ancestor and descendant tables add in size, so their exponent coefficients are maximized rather than summed.

Theorem 5.1 (Degree-51/100 specialization)**.** *Under the normalized input and entry-access premises of Section 2, strengthened to maximum initial degree at most* $\lceil s^{51/100} \rceil$*, there is a deterministic exact contraction algorithm with*

$$T \le \mathrm{poly}(s+\mathcal{L})\, 2^{(\widehat{C}_T + o(1))s/\sqrt{\lambda}},$$
$$S \le \mathrm{poly}(s+\mathcal{L})\, 2^{(\widehat{C}_S + o(1))s/\sqrt{\lambda}},$$

where

$$\widehat{C}_T = C_1 = 13.68968074357885\ldots, \qquad \widehat{C}_S = \frac{20}{21} C_0 = 12.79756505003982\ldots.$$

Thus 13.690 *and* 12.798 *are safe eventual coefficients for time and space.*

Proof. The strict gate table, depth bootstrap, and precision argument establish all premises of the imported local contraction and the new joint certificate at every active call. The fully charged recurrence and the strict branch comparisons give the time bound. The local space comparisons give the space bound. All remaining entry and arithmetic costs were included above. Exactness follows from the unchanged contraction identities and plan regrouping, independently of the parameter estimates. □

5.3 What causes this specialization's improvement

A same-schedule comparison is essential. Replacing the new root coefficient by the older pairwise certificate gives root coefficient 13.46705560666417.... That is still below C_1, so the old and new root certificates give *the same overall time coefficient* on this schedule. The low-degree time improvement is therefore a degree-aware composition of the available ingredients; it must not be attributed to the new root potentials. The new joint certificate separately supplies the smaller root space guarantee.

The old time-support argument on this schedule bounds local root space by 13.46705560666417...; the new joint argument bounds it by 12.79756505003982.... Dividing the old root bound by 21/20 would require its own augmented-support certificate and is not justified merely by the new selector's existence. These comparisons concern valid upper bounds, not matching lower bounds for actual memory use.

The standalone `check_low_degree.py` verifies the exact gate gaps, recurrence radicals, interval comparisons, and the unchanged dominant time branch for both root certificates. It also rejects three deliberately false claims, including reuse of this schedule at degree exponent 0.52. The specialized coefficients remain per normalized wire; converting them into a numerical coefficient per 3-SAT variable requires the explicit wire ledger of the chosen reduction.

5.4 Earlier usable bounds and a stronger inherited comparator

The printed 25.163765... polynomial-space bound at degree $s^{0.51}$ is not the only fair comparator [22]. The eight-group cached proof in [23] uses, in its smaller-degree branch, $\eta_d = 1/(15 \cdot 2^d)$. Its root and retained-degree exponents are $\delta + 2/5$, while its fresh-side and capacity exponent is at most 14/15. It therefore works for every fixed $\delta < 3/5$, including 51/100, although the stated corollary uses the stronger $s^{1/12}$ premise. This already gives time 19.08387285... and space 16.99635053.... Replacing

its matrix allowance by the available 12/5 bound leaves its literal block-table construction as the bottleneck, with time coefficient $(\pi/\ln 2)\sqrt{135/8} = 18.61856916\ldots$. Omitting the table-construction cost would give an invalid comparison.

A closer comparator simply multiplies every taper in the incoming generic $10 \to 13 \to 7$ schedule by 17/4, keeping its old pairwise certificate. Its cumulative tapers are 5117/80000, 9163/80000, and 34/225. The degree and capacity gates hold for $\delta = 51/100$; the tail check uses the stricter certified bound $\log_2 7 < 14037/5000$. The dominant root coefficient is then

$$\frac{29.81160894538347\ldots}{\sqrt{17/4}} = 14.46075441781\ldots.$$

The exact checker includes these gates and all paid branches. This is an inherited composition evaluated here, not a theorem printed in the source. It is a substantially stronger baseline than comparing only with the old generic headline or the polynomial-space number.

Construction valid at degree $s^{0.51}$	Time	Space class
Printed exact-kernels branch	$25.16377\ldots$	Polynomial
Eight-group proof, extended premise	$19.08387\ldots$	Subexponential
Eight groups, faster backend, tables paid	$18.61857\ldots$	Subexponential
Old pairwise schedule scaled by 17/4	$14.46075\ldots$	Subexponential
New stage-specific schedule	$13.68968\ldots$	Subexponential

The last improvement is about 5.33% relative to the immediately preceding time coefficient. The new root potentials are not its cause, as the same-schedule calculation above makes explicit. None of these comparisons proves optimality among all retunings of older primitives. For the stronger degree-$s^{1/12}$ promise, earlier cached coefficients as small as $13.37965\ldots$ remain available; the present specialization does not supersede that separate regime [10].

6 Consequences for 3-SAT and what remains unresolved

6.1 A strict generic coefficient gain with the same asymptotic scale

Under the generic $s^{9/10}$ degree promise, the fastest comparable incoming time coefficient was $C_{\mathrm{old}} = 29.81160894538347\ldots$. Theorem 4.1 decreases it by $0.06555185158\ldots$, approximately 0.21989%. Relative to that packet's accompanying space upper bound, the new space coefficient drops by approximately 4.97%. These percentages describe coefficients in exponential upper bounds, rather than measured speedups or memory ratios.

If $E(s) = s/\sqrt{\lceil \log_2 s \rceil}$, the leading expressions compare as

$$\frac{2^{C_{\mathrm{old}} E(s)}}{2^{C_T E(s)}} = 2^{(C_{\mathrm{old}} - C_T)E(s)}.$$

Since the coefficient gap is fixed and positive, it is eventually larger than the difference of the respective $o(1)$ terms. This is why the change is a strict asymptotic improvement in the quantitative certificate despite preserving the exponent's logarithmic order. No lower bound on the older implementation follows from this ratio.

The useful resource alternatives are retained. The affine packet gives $(30.2570356\ldots, 25.9890677\ldots)$, with more time and less space. The generic polynomial-space construction has time coefficient $46.1848456\ldots$. Our cached result is not a replacement for either resource requirement [14, 10].

6.2 Transfer to the actual 3-SAT reductions

Counting assignments also decides satisfiability by testing whether the count is nonzero. We restrict all applications here to clauses of width at most three. The stronger counting interface is useful because the inherited reductions reconstruct the exact count.

For fixed clause density, the native variable–clause network has $O_D(n)$ wires. High-degree variables can be conditioned or represented by the exact bounded-degree equality constructions already used in the corpus. Invoking the improved contractor preserves the established order

$$(n+m)^{O(1)}2^{O_D(n/\sqrt{\log n})}.$$

At arbitrary density, the source's sparse parity tests and incidence encoding produce $O(n \log\log n)$ wires, so the order remains

$$(n+m)^{O(1)}2^{O(n\log\log n/\sqrt{\log n})}.$$

In both cases the appropriate reduction must supply the degree, precision, entry-access, and number-of-branches promises. The normalized coefficient C_T cannot be read as a coefficient per SAT variable without explicitly multiplying by the reduction's wire constant and charging its other work.

There is a more precise ledger for the arbitrary-density construction. Write $\lambda_n = \lceil \log_2 n \rceil$ and fix a constant $c \geq 1$ for the source's sparse-query guarantee: at most $a \leq 6n$ equations, each of degree at most three with at most λ_n^c monomials. Then

$$K = 3(a+2n) \leq 24n, \qquad N \leq 3a\lambda_n^c \leq 18n\lambda_n^c.$$

The grouping in Section 7 has $B = o(n)$ nonempty group pairs and degree $\sqrt{n}$ polylog(n). Its wire bound gives

$$\begin{aligned} M &\leq B + \log_2 \binom{N+K}{K} \\ &\leq (24c + o(1))\, n \log_2 \lambda_n. \end{aligned}$$

The corresponding integer upper budget covering these wires and $n+1$ recovery precision is computable from the displayed binomial bound and satisfies $s \leq (24c+o(1))n\log_2\lambda_n$ and the degree-$s^{51/100}$ promise eventually. It yields the parametric leading contraction coefficient $24c\,\widehat{C}_T$ on $n\log_2\lambda_n/\sqrt{\lambda_n}$. This is an explicit ledger in the fixed query parameter c; we have not audited a numerical value of c through the whole parity construction.

The outer work does not introduce another leading coefficient at this scale. Sparsification has $2^{O(n/\sqrt{\lambda_n})}$ nodes, high-incidence branching has at most $2^{n/\sqrt{\lambda_n}}$ branches, the improved array count is $2^{O_c(n/(\sqrt{\lambda_n}\log\lambda_n))}$, and parity construction and candidate enumeration cost $2^{\lambda_n^{O_c(1)}}$ beyond the contractions. Their logarithms are all $o(n\log\lambda_n/\sqrt{\lambda_n})$ [4]. Thus the smaller normalized coefficient improves the contraction contribution using the same fixed 3-SAT reduction. It does not make 13.690 a numerical coefficient per original SAT variable.

Where the stronger degree-$s^{1/12}$ promise is available, the earlier 13.37965… cached option remains preferable to this specialization. The present result is not a domination statement for every density regime.

The density-aware packets additionally retain the bound

$$(n+m)^{O(1)}2^{O\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}$$

in their stated regime. For example, at $m = \Theta(n \log\log n)$ they replace a $\log\log n$ numerator by $\log\log\log n$. This genuine regime-dependent order improvement is prior work and is not attributed to the present certificate [15, 16].

6.3 Why the present mechanism cannot supply the requested order gain

The finite norm certificate affects only a constant in the inherited local scale $M/\sqrt{u}$. The gate conditions retain $u = \Theta(\log s)$. Changing support weights or finite menus therefore preserves $M/\sqrt{\log s}$ unless another part of that interface changes. The root precision cannot simply be reduced to a small constant: the Hermite recovery increment still has size $\Theta(MD/u)$.

Removing the general-case $\log\log n$ penalty requires an encoding with $o(n \log\log n)$ wires and paid local entries, or another method for summing the incidence information. Replacing $\sqrt{\log n}$ by $\log n$ requires a different contraction or mode-summation mechanism. The earlier precision packet explicitly labels its structured-mode acceleration as unproved [17]. Its conditional bound must not be presented as an available algorithm, or ranked numerically against the proved coefficient here.

The next section tests two concrete approaches to that larger goal. The result is a representation barrier and a tighter enumeration bound, rather than an asserted solution of the missing algorithmic step. The small remaining freedom in the fixed nine-plan average, quantified in Section 3.5, also argues against spending the next iteration on reweighting those same plans alone.

7 Structural limits of two routes to a larger improvement

The new consistency certificate improves a leading contraction coefficient. Two tempting ways to improve the logarithmic order are to compress the variable information further and to lower the initial tensor degree. This section gives precise limits of particular implementations of those ideas. None is a lower bound on the running time of 3-SAT.

7.1 The variable-union pairing has full rank

In the incidence encoding of the source reduction [4], the variable-side tensor depends only on whether a variable appears in a selected incidence monomial. If S is its set of touched variables, summing that monomial over Boolean assignments gives $2^{n-|S|}$. It is therefore natural to ask whether a small algebra can retain just the information needed to combine such sets.

Theorem 7.1 (A fixed linear summary requires exponential dimension)**.** *Let $\mathbb{F}$ be any field and put*

$$A_n = \mathbb{F}[x_1, \dots, x_n]/(x_i^2 - x_i : 1 \le i \le n).$$

For $S \subseteq [n]$, write $e_S = \prod_{i \in S} x_i$, and define the linear functional $L : A_n \to \mathbb{F}$ by $L(e_S) = 2^{n-|S|}$, with the integer interpreted in $\mathbb{F}$. Suppose $E : A_n \to V$ is linear, $\dim_{\mathbb{F}} V = d$, and a bilinear form $b : V \times V \to \mathbb{F}$ satisfies

$$b(E(f), E(g)) = L(fg) \qquad (f, g \in A_n).$$

Then $d \ge 2^n$.

Proof. The squarefree monomials e_S form a basis and satisfy $e_S e_T = e_{S \cup T}$. Consequently the matrix

of the pairing $(f, g) \mapsto L(fg)$, in this basis, is

$$H_{S,T} = 2^{n-|S\cup T|} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{\otimes n}_{S,T}. \tag{U}$$

Order subsets by their binary indicator strings. Each coordinate contributes 2 when both membership bits are zero, and 1 otherwise, proving the Kronecker identity. The one-coordinate matrix has the integer inverse

$$\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Thus H is invertible over every field, including characteristic two, and has rank 2^n. On the other hand, if Q is the matrix of E and B the matrix of b, then $H = Q^{\mathsf{T}} B Q$ has rank at most d. □

This statement also applies to a linear encoding into a finite-dimensional algebra whose multiplication is followed by a linear readout: their composition is such a bilinear form. The basis inputs already arise from 3-CNFs of width at most three, since the conjunction of the positive unit clauses indexed by S has satisfaction polynomial e_S.

There is an equally direct decision version. For each $\alpha \in \{0,1\}^n$, let F_α be the conjunction of unit clauses fixing the assignment to α. Then

$$\left(\mathbf{1}_{\mathrm{SAT}(F_\alpha \wedge F_\beta)}\right)_{\alpha,\beta} = I_{2^n}.$$

Any fixed d-dimensional summaries whose conjunction decision is obtained by a bilinear form over a field therefore require $d \geq 2^n$, whether or not the summary assignment is linear.

These are constraints on a fixed compositional representation. They do not exclude adaptive restrictions, special reachable families, nonlinear evaluation, or another summation method. In particular, they do not show that the actual sparse parity-query family needs all 2^n dimensions. They explain why merely identifying incidence subsets with variable unions does not by itself supply the missing efficient algorithm.

7.2 Budget-aware counting in the existing grouping reduction

The source groups tests and variables separately into $J = \lceil n^{1/4} \rceil$ groups [4]. Increasing J to nearly $\sqrt{n}$ is already present in the corpus: the recursive refinement uses $J = \lceil n^{0.495} \rceil$ [19], and the high-moment packet uses $J = \lceil \sqrt{n} / \lceil \log_2 n \rceil^2 \rceil$ [18]. The latter explicitly obtains initial degree $\sqrt{n}\,\mathrm{polylog}(n)$. We credit that result; the refinement below concerns its array-counting overhead.

Consider the source's sparse-query interface with degree at most three, $a \leq cn$ tests and at most λ^c monomials per test, for fixed c, where $\lambda = \lceil \log_2 n \rceil$. After its high-incidence branching, each test has at most $3\lambda^c$ incident variables, each free variable has $O_c(\lambda^{c+1/2})$ incidences, and the number of branches is at most $2^{n/\sqrt{\lambda}}$. The retained incidence-degree bound is $K = 3(a + 2n) = O_c(n)$.

Partition both sides into J groups balanced in cardinality. Let N_{gh} be the incidence count for a group pair, let $N = \sum_{g,h} N_{gh}$, and let B count the nonempty pairs. Thus $B \leq J^2$ and $N = O_c(n\lambda^c)$. The enumerated arrays satisfy

$$0 \leq d_{gh} \leq N_{gh}, \qquad \sum_{g,h} d_{gh} \leq K.$$

Lemma 7.2 (Array and wire budgets)**.** *The number R of such arrays satisfies*

$$R \le \min\left\{(K+1)^B,\ \binom{K+B}{B},\ \prod_{N_{gh}>0}(N_{gh}+1)\right\}. \tag{A}$$

For each array the source's subset-index encoding has wire count M and initial maximum degree Δ_0 satisfying

$$M \le B + \log_2 \binom{N+K}{K}, \qquad M \le N,$$
$$\Delta_0 = O_c((n/J+1)\lambda^{c+1/2}).$$

Its local-entry evaluation still takes $\mathrm{poly}(n)2^{\lambda^{O_c(1)}}$ *time and polynomial space.*

Proof. The three bounds in (A) respectively discard the sum constraint, discard the individual capacities and count weak compositions of a sum at most K, and discard only the sum constraint while retaining the capacities. For a fixed array, pair (g,h) uses $\lceil \log_2 \binom{N_{gh}}{d_{gh}} \rceil$ wires. Their product of subset counts is bounded by the number of monomials of total degree at most K in N indeterminates, namely $\binom{N+K}{K}$. Summing the rounding errors gives the first wire bound. The inequality $\lceil \log_2 \binom{t}{j} \rceil \le t$ gives the second.

A group vertex has degree at most its incidence count; combine the balanced group sizes with the test and variable incidence bounds. A test-group entry is a product of independent per-test coefficients. Each is evaluated by the source's finite-difference formula on one polylogarithmic scope. The times for these evaluations add; they do not multiply. The variable-group entry is the power of two recording untouched variables. Subset decoding, ring arithmetic and all descriptions have polynomial size. □

The composition count improves the allowable polylogarithmic choice of J. For $J \asymp \sqrt{n}/\lambda^A$ with fixed $A > 0$, it gives

$$\log_2 R \le B \log_2\left(\frac{\mathrm{e}(K+B)}{B}\right) = O_c\left(\frac{n\log\lambda}{\lambda^{2A}}\right).$$

Hence $A > 1/4$ suffices for $o(n/\sqrt{\lambda})$ overhead; the crude $(K+1)^{J^2}$ estimate would require $A > 3/4$. For an exact integer choice, put $\nu = \lceil \log_2 \lambda \rceil$ and take

$$J = \max\{j \in \mathbb{Z}_{\ge 0} : j^2 \lceil \sqrt{\lambda} \rceil \nu^2 \le n\}.$$

Treat the finite range with $J < 1$ directly. Then

$$J = \Theta\left(\frac{\sqrt{n}}{\lambda^{1/4}\nu}\right), \quad \log_2 R = O_c\left(\frac{n}{\sqrt{\lambda}\,\nu}\right), \quad \Delta_0 = O_c(\sqrt{n}\,\lambda^{c+3/4}\nu).$$

The wire bound remains $O_c(n\nu)$, and $B = o(n)$. Arrays can be generated by a depth-first search with a remaining-degree counter, using polynomial work per array and polynomial space. The extra exponent outside contraction is $O(n/\sqrt{\lambda}) + O_c(n/(\sqrt{\lambda}\,\nu))$, including the original high-incidence branching.

Thus the improvement is in polylogarithmic slack, while the initial degree exponent remains $1/2$. A schedule accepting $s^{1/2}(\log s)^{O(1)}$ degrees may use that bound directly. A theorem requiring exactly $\Delta_0 \le s^\gamma$ may instead use any fixed $\gamma > 1/2$, unless its parameter construction explicitly absorbs the polylogarithmic factor. The high-moment packet supplies such an integer budget [18]. This reasoning does not establish the stronger degree promise $s^{1/12}$ for arbitrary-density queries.

7.3 Small groups can force exponentially many degree arrays

The next result strengthens the warning against simply choosing many more groups. Its quantifiers allow the partition to be chosen after seeing the graph.

Theorem 7.3 (A graph-adaptive grouping obstruction)**.** *Fix $0 < \delta < 1/2$ and an integer d with $d(1-2\delta) > 3$. For every sufficiently large n there exists a bipartite graph with n vertices on each side and maximum degree at most d such that every partition of each side into groups of size at most n^δ has more than n nonempty group pairs.*

Proof. An ordered d-tuple of permutations of $[n]$ specifies d perfect matchings. Forget the colors and repeated edges; the resulting simple graph has maximum degree at most d. There are

$$(n!)^d = \exp(dn \ln n + O_d(n))$$

tuples. We upper-bound the tuples whose graph has a prohibited partition.

Label groups by numbers in $[n]$. There are at most n^{2n} pairs of labeled partitions. A support of at most n group pairs can be chosen in at most

$$(n+1)\binom{n^2}{n} = \exp(n \ln n + O(n))$$

ways. If groups have size at most $q = \lfloor n^\delta \rfloor$, this support allows at most $Q = nq^2 \le n^{1+2\delta}$ individual vertex pairs.

The projected graph has at most dn distinct edges. Its edge set has at most

$$\sum_{t=0}^{dn} \binom{Q}{t} \le \binom{Q+dn}{dn} = \exp(2d\delta n \ln n + O_d(n))$$

possibilities. The inequality follows by representing a subset of size at most dn as a multiset of size exactly dn after adjoining a dummy element. For a fixed graph of maximum degree at most d, at most d^{dn} permutation tuples project to it: at each left vertex each color chooses one of at most d neighbors. The permutation requirement only reduces that count.

Multiplying these bounds gives $\exp((3+2d\delta)n \ln n + O_d(n))$ bad tuples. This is less than $(n!)^d$ for large n by the assumed strict inequality. Some tuple therefore gives a graph with the stated property. □

Corollary 7.4 (The total-degree constraint does not remove this cost)**.** *For the graphs in Theorem 7.3, every permitted grouping has at least 2^n valid degree arrays whenever $K \ge n$.*

Proof. Choose any n nonempty group pairs, assign $d_{gh} \in \{0,1\}$ independently on them, and assign zero elsewhere. All capacities are respected and the total is at most n. □

For example, $\delta = 1/10$ and $d = 4$ satisfy the strict inequality. Group sizes $n^{1/12}\,\mathrm{polylog}(n)$ are eventually at most $n^{1/10}$. The result therefore blocks a direct attempt to obtain the $s^{1/12}$ promise by uniformly shrinking all groups while continuing to enumerate the same arrays.

The graphs fit the generic sparse-query interface: take one polynomial per left vertex equal, over $\mathbb{F}_2$, to the sum of its neighboring variables. These have degree one and scopes of size at most d, so there is no high-incidence branching and $K \ge n$. This example is easy by Gaussian elimination. Its

role is precisely to show that the literal outer enumeration can be expensive even on easy inputs unless further algebraic structure is used.

Finally, small group cardinality is a *sufficient* way of guaranteeing small encoded degree, not a necessary one. A specific array with $d_{gh} = 0$ or N_{gh} can have zero wires on that pair even when the group is large. The theorem neither rules out exploiting these cases nor lower-bounds the number of arrays with nonzero coefficient. It rules out only the stated combination of uniformly small groups and exhaustive enumeration of combinatorially allowed arrays.

The accompanying `check_structures.py` verifies the integer Kronecker inverse, its reductions in characteristic two and odd characteristics, direct small Boolean counts, and capacity-constrained array counts against independent exhaustive enumeration. Deliberately corrupted matrices and counts must be rejected. These checks support the finite identities; the quantified graph obstruction follows from the proof above.

8 Verification, reproduction, and proof status

8.1 Exact certificates and independent reconstruction

All acceptance decisions use integer or rational arithmetic. The primary checker and an independent checker reconstruct the same support families from the ordered block partitions using different code. The checks include:

- all 648 time-family conservation identities, all 186624 pair choices, exact maxima, and the displayed rational β_{10}^2;
- all 801 augmented conservation identities and 98532 additional mixed time/boundary choices for the joint selector;
- strict packing margins, the exact taper sum, all controlling time branches, the root storage maximum, and rational enclosures proving the safe eventual coefficients;
- the degree-51/100 specialization, its stage-specific gates, the same-schedule comparison with the old certificate, and a uniformly scaled inherited comparison schedule;
- all 35937 tuples of the inherited thirteen-group certificate and the 15 supports of the seven-group tail;
- the exact Gram matrix, positive LDL pivots, stationary positive weights, the attained upper certificate, and its gap from the mixture lower bound;
- the Kronecker inverse and capacity-array identities in Section 7, checked against independent small exhaustive computations.

The coefficient enclosures use the alternating series for arctan in Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, a positive geometric-tail bound for the series of $\ln 2 = 2\,\mathrm{artanh}(1/3)$, and integer-square-root enclosures. Decimal output is for readability and does not decide an inequality.

Deliberate negative controls corrupt a potential, an ordered plan, a claimed bound, a Gram entry, stationary weights, and the structural identities. A larger storage multiplier 6/5 is rejected by the fixed certificate. The checks raise explicit exceptions, so running Python with optimization cannot silently disable them.

8.2 A finite exact contraction prototype

The prototype evaluates the graphical sum over $(\mathbb{Z}/64\mathbb{Z})[\epsilon]/(\epsilon^2)$, which includes zero divisors and nilpotents. On 32 seeded fixtures it compares all nine regroupings against an independent exhaustive sum, for 288 comparisons. Eighteen reference answers are nonzero. It checks every plan's exact number of paid singleton-factor evaluations, then checks the selected joint time/storage plan against the same reference. Empty index lists and invalid domains are separate controls.

The fixtures use power-of-two domain sizes, so their selector evaluates the integer support scores on exact integer logarithms. The mathematical selector for arbitrary integer list sizes is the product comparison in Section 2. The prototype uses ordinary scalar multiplication; it does not implement the asymptotic rectangular backend. These are correctness checks of a finite kernel, not SAT runtime measurements.

8.3 Reproduction commands and files

Python 3.10 or later and its standard library suffice for verification. From the extracted archive directory run

```
python3 code/verify_manifest.py
python3 code/run_checks.py
python3 code/build_report.py
```

The first command checks the shipped bytes. Run it before rebuilding: PDF metadata can change on a later build. The second runs eight exact checkers both normally and with `-O`, requires identical parsed outputs, and writes separate receipts under `replay_results/`. All sixteen executed checks passed for the delivered package. The third requires a normal pdfLaTeX installation and the packages named in the preamble. It disables shell escape and rejects overfull boxes or unresolved references.

The optional `discover_certificate.py` uses NumPy to repeat the finite potential search for the frozen rational weights. It is unnecessary for verification; the exact certificate is the witness being checked. `CLAIMS.json` distinguishes newly proved results, finite checks, imported results, and unresolved questions.

8.4 Limits of the verification claim

No full fast SAT solver, fast-matrix implementation, new Lean proof, or fresh kernel replay of the two source developments is included. The new local inequalities are exact finite theorems with complete witnesses; the asymptotic contraction theorem imports the analytic and arithmetic interfaces explicitly stated earlier. The source inventory and hashes establish which bytes were reviewed, not the truth of every external theorem. The quantified representation and graph obstructions follow from their mathematical proofs; their small checks are supporting evidence.

The delivered result is therefore a reproducible quantitative refinement with precise dependencies. The unrestricted exponent-order improvement asked for remains an open task in this research iteration.

A Review inventory and source coverage

The frozen snapshot contains twenty-nine filed refinement packets and five new incoming archives. The two primary manuscripts are additional sources, not part of this count. Every packet's title, headline claims, and available status material were screened. This identifies the claimed comparison frontier; it is not an independent proof audit of every packet.

Detailed examination focused on the exact mode and precision interface, asymmetric packing, adaptive flags, complete contraction trees, the rectangular backend, and the strongest compatible incoming bounds. The verification section distinguishes executable finite checks from imported analytic theorems and source inspection of the Lean projects.

The tables record author-stated contributions and claim scope. A coefficient refinement preserves the logarithmic exponent order relative to imported identities. A conditional stronger order requires its named missing interface. Kernel measurements are not full SAT measurements. Generic coefficients refer to the maximum-degree promise $\Delta \leq \lceil s^{9/10} \rceil$; smaller-degree bounds must be compared separately, including where an actual 3-SAT reduction supplies that stronger promise. Each full identifier links to the frozen directory or archive. All applications in this report are to 3-SAT, even where older packets treat broader clause widths.

Filed round 1: seven packets

Packet identifier	Main contribution and claim status
cauchy_mode_improvement	Integral factorial Cauchy modes reduce the mode-count coefficient. Coefficient refinement; no smaller unrestricted exponent order.
eth_contraction_improvements	Exact Hermite taper, smaller precision budget, and streamed packet extraction. Coefficient refinement with an exact local prototype.
3sat_precision_refinement	Sharper precision and a solved taper minimax problem. Its stronger logarithmic order requires an explicitly unproved structured-mode algorithm.
sat_density_improvement	Density-aware counting improves a growing-density regime. General worst-case and fixed-density exponent orders remain unchanged.
eth_density_refinement	Independent density-sensitive refinement and smaller precision allowances. Same density-regime gain; unrestricted order unchanged.
3sat_refinement	Scalar parity lift reduces incidence degree. A second contraction needs an unsupplied compact factorization; its stronger order is conditional.
3sat_asymptotic_analysis	Scoped taper, budget, and Newton-mode barriers. Two missing interfaces are identified; no stronger unrestricted bound is proved.

Filed round 2: eight packets

Packet identifier	Main contribution and claim status
eth_asymmetric_packet_balancing	Asymmetric pair-load balancing, nearly lossless padding, and a precision buffer. Coefficient refinement; degree-sensitive branches have stronger premises.
3sat_round2_refinement	Matching-potential packing, thin precision gaps, and a revised schedule. Coefficient refinement with an independent review and scoped rank barriers.
eth_balanced_packing	Deterministic two-stage labeling and an additive precision guard. Polynomial-space coefficient refinement; no practical crossover claim.
eth_high_moment	Higher-moment labeling and degree-sensitive schedules. Square-root-degree grouping with polylogarithmic slack is already present here.
eth_matching_refinement	Exact rational matching potential and thin additive gaps. Independently reaches the same generic coefficient as two sibling packets.
eth_packet_refinement	Sharper pair-load packing, near-unit padding, and an additive guard. Coefficient refinement; logarithmic exponent order unchanged.
eth_recursive_refinement	Residue-aware Cauchy basis and smaller recursive precision cost. Coefficient refinement; larger test/variable group counts are already explored.
eth_joint_cauchy	Full crossing-block Cauchy modes and cached group contractions. Separate polynomial-space and cached-space coefficient guarantees.

Filed round 3: five packets

Packet identifier	Main contribution and claim status
sat_adaptive_contraction	Adaptive streamed separators and cached nine-group aggregation. Coefficient refinements and modular Cauchy kernel measurements; no full SAT timing.
sat_frontier_improvements	Combined schedules, degree-sensitive branches, and quadratic feature tables. Coefficient and kernel refinements; resource assumptions remain distinct.
subexponential_sat_frontier	Faster exact labeling and precision peeling with a revised schedule. Coefficient refinement and measured subroutine speedups only.
subsat_contraction_tree	Cached subset tables, a paid tree, and an integer/CRT matrix bridge. Cached coefficient refinement; fast matrix backend is theoretical.
subexponential_sat_kernel_improvements	Compatible schedule compositions, adaptive crossing separators, and streamed/projective extraction. Exact kernel improvements; no order gain.

Filed round 4: nine packets

Packet identifier	Main contribution and claim status
sat_adaptive_bounds	Adaptive complete binary flags. Generic degree-$s^{9/10}$ polynomial-space coefficient 46.18485 . . .; stronger low-degree branches are separate.
sat_adaptive_tree_certificate	Adaptive complete contraction trees with exact supports. Former generic cached frontier 30.89102 . . . /30.75391 . . ..
sat_adaptive_frontier	Adaptive quartet conditioning and joint cached optimization. Coefficient refinement; no smaller unrestricted exponent order.
sat_adaptive_matrix_partition	Balanced sixteen-group separator optimization and an adaptive triangle partition. Coefficient refinement with finite dynamic-programming certificates.
subexponential_sat_adaptive_four	Four-group matching inequalities and adaptive cached contraction. Coefficient refinement; matrix exponent allowances are stated separately.
subsat_rectangular	Rectangular matrix contraction, streamed conditioning, and precision barriers. Supplies the backend reused in the incoming pairwise packet.
eth_adaptive_exact_kernels	Degree-sensitive schedules, sparse-support scoring, and fused Cauchy transitions. Includes 25.16377 . . . polynomial-space time at degree $s^{0.51}$.
3sat_round3_frontier	Low-degree schedules and centered extraction filters. Its earlier degree-$s^{1/12}$ polynomial-space coefficient is superseded by the pairwise arrival.
eth_adaptive_schedules	Adaptive schedules and an incremental matching-potential kernel. Coefficient and local complexity improvements; no full SAT solver.

Five newer incoming packets

Packet identifier	Main contribution and claim status
sat_affine_cauchy	Affine directions, cyclic menus, and guarded fast Cauchy transforms. Generic cached coefficients 30.41335 . . . /29.46603 . . ..
sat_affine_contraction	Affine plans with joint time–space selection. Generic cached coefficients 30.25704 . . . /25.98907 . . .; smaller-memory alternative.
sat_affine_geometry	Affine and punctured menus with paid recursion. Generic cached coefficients 30.26987 . . . /29.48754 . . .; exact Toeplitz prototypes.
sat_affine_plane	Affine-line partitions and a thirteen-group certificate. Generic cached coefficients 30.69134 . . . /29.46603 . . .; order unchanged.
sat_pairwise_certificates	Weighted pair-support maxima and rectangular contraction. Prior generic cached coefficient 29.81161 . . .; degree-$s^{1/12}$ polynomial-space coefficient 16.81224 . . ..

The downloaded-source inventory records paths, Git blob identifiers, sizes, and SHA-256 hashes. Hash agreement establishes source identity, not proof validity. Later repository changes do not alter this frozen comparison.

References

[1] V. Reshetnikov, *ProveIt: Sub-exponential 3-SAT*, project index and trust-status statement, frozen October 26, 2026. Pinned project index.

[2] ProveIt, *New research reports*, commit `26bd83d4b37d3c5a3ccb45bab2e6385fae3977f1`, October 26, 2026, 8:51 PM Pacific. Five-packet arrival commit; frozen incoming directory.

[3] ProveIt, *Round-four placement and public formalization link*, 2026. Placement commit a7ae5ab55f96; formalization-link commit 7de9d09fecd1.

[4] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026. The unmodified vendored manuscript and its quantitative appendices are explicit dependencies. Pinned manuscript directory; quantitative contraction source; quantitative reduction source.

[5] OpenAI, *Subexponential SAT formalization scope*, document 378, commit `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. Pinned Lean scope statement. Its displayed exponent has the additional logarithmic factor discussed in Section 1.

[6] Anthropic, *A $2^{o(n)}$ Time Algorithm for 3-SAT: formal statements*, `eth/MainResults.lean`, commit `1a6eee293b9e00cb432eef6018a21f523623a9af`, 2026. Exact theorem statements; manuscript at the same commit.

[7] Anthropic, *Proof entry points for the 3-SAT development*, `eth/Solution.lean`, same exact commit. Proof entry points; project instructions; user-specified pull-request revision.

[8] *Asymmetric packet balancing for subexponential SAT*, ProveIt research packet, October 26, 2026. Pinned report source. Used for asymmetric pair loads, nearly lossless padding, fresh-side degree accounting, and the additive precision buffer.

[9] *Full Crossing-Block Cauchy Modes*, ProveIt research packet, October 26, 2026. Pinned report source. Used for the integral mode weights and cached factor representation.

[10] *Adaptive contraction beyond the October 26 SAT frontier*, ProveIt research packet, October 26, 2026. Pinned report sections; adaptive flag theorem.

The separate cached lower-degree theorem is in the cached-contraction section.

[11] *Adaptive Contraction Trees with Exact Support Certificates*, ProveIt research packet, October 26, 2026. Pinned report source. Used for complete operation supports, the paid cache recurrence, and the thirteen- and seven-group certificates.

[12] *A Smaller Explicit Contraction Bound and Stronger Precision Barriers*, ProveIt research packet, October 26, 2026. Pinned rectangular-contraction section; complete report. Used for rectangular interpolation, the coordinate/CRT bridge, and the workspace contract.

[13] *Pairwise Certificates for Adaptive SAT Contraction*, incoming research packet, October 26, 2026. Pinned complete archive. The principal generic comparison: nine ten-group plans, integer rectangular supports, weighted pair maxima, and the strict $10 \to 13 \to 7$ schedule. Also states a low-degree polynomial-space improvement and scoped separator/crossing obstructions.

[14] *Affine-Plane Contraction Plans: Smaller Exponent Coefficients and a Joint Time–Space Certificate for the Exact SAT Contraction Framework*, incoming research packet, October 26, 2026. Pinned complete archive. The smaller-memory generic comparison and prior simultaneous boundary-support selection argument.

[15] *Improving the Clause-Density Dependence of Exact k-SAT Counting*, ProveIt research packet, October 26, 2026. Pinned report source. Only its width-three specialization is used here.

[16] *A density-sensitive refinement of the October 2026 exact #k-SAT counting algorithm*, ProveIt research packet, October 26, 2026. Pinned report source. Independent density-regime refinement, with its stated density hypotheses.

[17] *Sharper precision for subexponential 3SAT: a proved schedule refinement and a conditional route to a better logarithmic exponent*, ProveIt research packet, October 26, 2026. Pinned report source. The stronger-order claim is explicitly conditional on a missing structured-mode algorithm.

[18] *Higher moments and economical precision gaps for subexponential SAT*, ProveIt research packet, October 26, 2026. Pinned report source. Prior source for square-root-degree grouping with polylogarithmic slack.

[19] *Refining the ETH contraction: a residue-aware Cauchy basis, sharper packing, and a cheaper recursive precision schedule*, ProveIt research packet, October 26, 2026. Pinned report source. Prior source for larger test/variable group counts.

[20] V. Vassilevska Williams, Y. Xu, Z. Xu, and R. Zhou, *New Bounds for Matrix Multiplication: from Alpha to Omega*, SODA 2024; arXiv:2307.07970v2, 2023. Primary paper. Its bounds $\omega \leq 2.371552$ and $\alpha \geq 0.321334$ suffice for the weaker rational allowances used here; no claim about the latest matrix exponents is required.

[21] M. J. Wainwright, T. S. Jaakkola, and A. S. Willsky, *MAP estimation via agreement on (hyper)trees: Message-passing and linear programming*, arXiv:cs/0508070v1, 2005. Primary paper. Background for decomposed upper bounds, agreement constraints, and their linear-programming/message-passing interpretation. Our exact inequality and certificate are given in Section 3.

[22] *Adaptive exact kernels and degree-sensitive schedules*, ProveIt research packet, October 26, 2026. Pinned report source. The printed degree-$s^{0.51}$ polynomial-space comparison is used here.

[23] *Further improvements beyond the October 26 SAT frontier*, ProveIt research packet, October 26, 2026. Pinned eight-group cached proof. Its smaller-degree schedule extends to degree exponent below 3/5 by the packing calculation in Section 5.